$$S = k \log W$$

which expresses (missing) information as the logarithm of the number of alternatives, Shannon's key idea. Boltzmann pointed out that this quantity coincides with the entropy used in thermodynamics. Entropy is 'missing information', that is, information with a minus sign. The total amount of entropy can only increase, because information can only diminish.[fn50]

Today, physicists commonly accept the idea that information can be used as a conceptual tool to throw light on the nature of heat. More audacious, but defended today by an increasing number of theorists, is the idea that the concept of information can be useful also to the mysterious aspects of quantum mechanics illustrated in Chapter 5.

Remember that a key result of quantum mechanics is precisely the fact that information is finite. The number of alternative results that we can obtain measuring a physical system[fn51] is infinite in classical mechanics; but, thanks to quantum theory, we have understood that, in reality, it is finite. Quantum mechanics can be understood as the discovery that information in nature is always *finite*.

In fact, the entire structure of quantum mechanics can be read and understood in terms of information, as follows. A physical system manifests itself only by interacting with another. The description of a physical system, then, is always given in relation to another physical system, the one with which it interacts. Any description of a system is therefore always a description of the *information* which a system has about another system, that is to say, the *correlation* between the two systems. The mysteries of quantum mechanics become less dense if interpreted in this way, as the description of the information that physical systems have about one another.

The description of a system, in the end, is nothing other than a way of summarizing all the past interactions with it, and using them to predict the effect of future interactions.

The entire formal structure of quantum mechanics can in large measure be expressed in two simple postulates:[1]

1. The relevant information in any physical system is finite.
2. You can always obtain new information on a physical system.

Here, the 'relevant information' is the information that we have about a given *system* as a consequence of our past interactions with it: information allowing us to predict what will be the result for us of future interactions with this system. The first postulate characterizes the granularity of quantum mechanics: the fact that a finite number of possibilities exists. The second characterizes its indeterminacy: the fact that there is always something